

• periodična struktura $\rightarrow$ osh. celica, ki se podobenkrat ponovi



-3 linearno neodv. vekt. $\vec{a}, \vec{b}, \vec{c}$

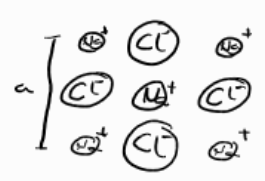
npr. $\vec{a} \perp \vec{b}, \vec{a} \perp \vec{c}, \vec{b} \perp \vec{c}, a=b=c \Rightarrow$ kubična mreža

$\rightarrow \vec{a} \not\perp \vec{b} \not\perp \vec{c}, a \neq b \neq c \Rightarrow$ triklinska mreža

• ionska vez

• lastnosti: stabilni, visoko tališče, prozorni, absorbirajo IR, dielektričnost

Zgled: vezana energija Na v NaCl



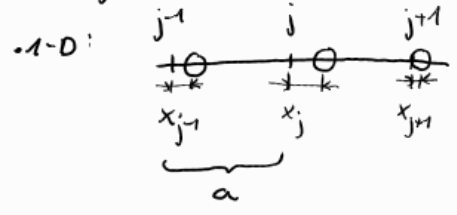
$$W_{Na^+} = \frac{e^2}{4\pi\epsilon_0(\frac{a}{2})} \left(-6 + \frac{12}{\sqrt{2}} - \frac{8}{\sqrt{3}} + \dots \right) = -\frac{\alpha_M e^2}{4\pi\epsilon_0 r}$$

$$\alpha_M = 1.75$$

(Madelungova konst.)

$$\rightarrow W = \frac{N}{2} W_{Na^+}$$

Nihanje kristalne rešetke



• N atomov

• K - konstanta vzmeti

• M - masa atoma

$$\rightarrow M \ddot{x}_j = -K(x_j - x_{j-1}) - K(x_j - x_{j+1})$$

Robni pogoji: $x_0 = x_N$

$$\rightarrow \text{rešitev bo oblike } x_j = A \exp(i(kaj - \omega t))$$

a_j - oddaljenost od izhodišča

$$\rightarrow -M A \omega^2 \exp(i(kaj - \omega t)) = -K(e^{i(kaj - \omega t)} - e^{i(kaj-1 - \omega t)}) - K(e^{i(kaj - \omega t)} - e^{i(kaj+1 - \omega t)})$$

$$M \omega^2 e^{ika_j} = K(2e^{ika_j} - e^{ika(j-1)} - e^{ika(j+1)})$$

$$M \omega^2 = K(2 - e^{-ika} - e^{ika}) = 2K(1 - \cos(ka)) = 2K \cdot 2 \sin^2 \frac{ka}{2} = 4K \sin^2 \frac{ka}{2}$$

$$\boxed{M \omega^2 = 4K \sin^2 \frac{ka}{2}} \text{ Disperzijska enačba (za zvok: } \omega = kc)$$

$ka \ll 1$:

$$M \omega^2 \approx K \left(\frac{ka}{2} \right)^2$$

$$\rightarrow \omega = \sqrt{\frac{K}{M}} ka$$

$x_0 = x_N$:

$$A e^{-i\omega t} = A e^{i(kaN - \omega t)}$$

$$e^{ikaN} = 1$$

$$\rightarrow kaN = 2\pi n, \quad \boxed{k = \frac{2\pi n}{Na}, n \in \mathbb{Z}}$$

1D: N nihanj

3D: 3N nihanj, 1x longitudinalno in 2x transverzalno

Notranja E in spec. topl. kristala

$$W_j = \hbar \omega (n_j + \frac{1}{2}), \quad n_j \in \mathbb{N} \cup \{0\}$$

• za vsako nihanje kanonična porazdelitev, $P(n_j) = \frac{1}{Z} e^{-\hbar \omega \beta n_j}$; $\beta = \frac{1}{k_B T}$ in $Z = \sum_{n_j} e^{-\hbar \omega \beta n_j}$

$$\bar{W}_j = \sum_{n_j=0}^{\infty} P(n_j) \hbar \omega n_j = \hbar \omega \cdot \frac{1}{(e^{\hbar \omega \beta} - 1)} = \hbar \omega_j \bar{n}_j$$

$$\rightarrow W_{\text{not}} = \sum_j \bar{W}_j = \sum_j \frac{\hbar \omega_j}{e^{\hbar \omega_j \beta} - 1} = \frac{3V \hbar}{2\pi^2 v_s^3} \int_0^{\omega_D} \frac{\omega^3}{e^{\hbar \omega \beta} - 1} d\omega$$

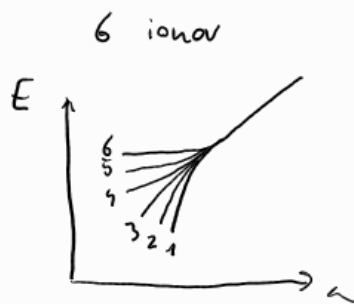
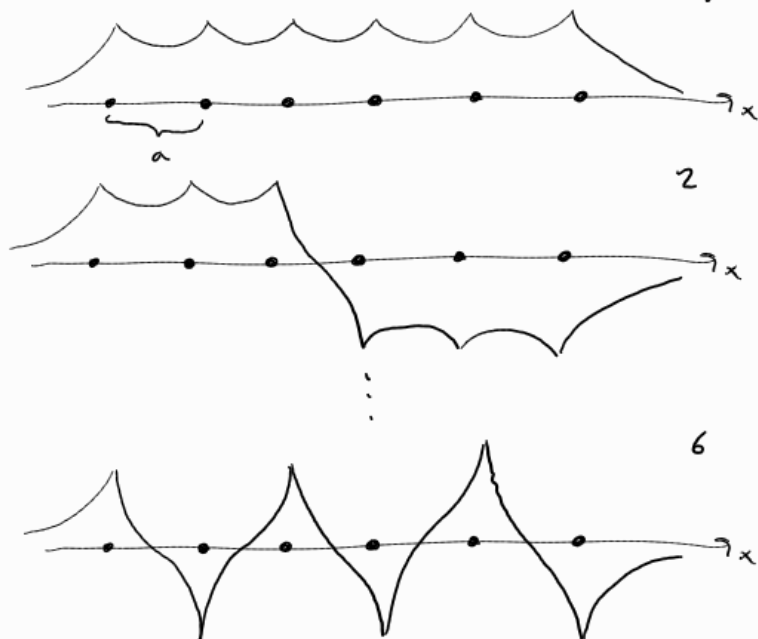
izpeljava STD

$$\rightarrow T \ll T_D: c_v \propto T^3$$

$$T \gg T_D: c_v = \text{konst.} = 3 \frac{m}{M} R$$

Elektroni v kristalih

1 (analogno z molekulo H_2^+ (MF1))



↓
kristal ($\sim N_A$ ionov)



→ iščemo E_e , $E = E(k)$

PRIBLIŽEK TESTNE VEZI

$$\Psi(x) = \sum_{j=1}^N c_j \underbrace{\phi(x - ja)}_{\phi_j} = \sum_{j=1}^N c_j \phi_j$$

energija za izoliran e^-

$$W_1 c_j - J(c_{j-1} + c_{j+1}) = W c_j; \quad J = - \int V_{j+1} \phi_{j+1}^* \phi_j dx$$

$$c_j = A e^{ikaj}$$

$$W_1 - J(e^{-ika} + e^{ika}) = W$$

$$W = W_1 - 2J \cos(ka)$$

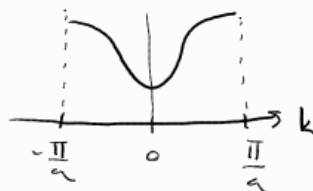
• robni pogoj: periodičnost ($\psi_0 = \psi_N$)

$$1 = e^{ikaN} \Rightarrow k_n = \frac{2\pi n}{aN}$$

$$-\frac{N}{2} < n < \frac{N}{2} \rightarrow \underbrace{-\frac{\pi}{a} < k < \frac{\pi}{a}}_{\text{Brillouinova zona}}$$

$\rightarrow ka \ll 1$:

$$W = W_1 - 2J(1 - \frac{1}{2}k^2a^2) = W_1 - 2J + Jk^2a^2$$



Prost e^- : $W = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}$

$$\rightarrow J \frac{k^2 a^2}{2m^*} = \frac{\hbar^2 k^2}{2m^*} \rightarrow \boxed{m^* = \frac{\hbar^2}{2Ja^2}} \text{ (efektivna masa)}$$

Kronig-Penneyev model kristala: potenciali so δ funkcije:



$$\cos ka = \cos k_0 a + \frac{mA}{\hbar k a} \sin k_0 a \text{ (izpeljava vaje)}$$

$$k = \sqrt{\frac{2mE}{\hbar^2}}$$

$$V(x - ja) = A\delta(x - ja)$$

$$\rightarrow ka, k_0 a \ll 1: \sqrt{1 - \frac{1}{2}k^2 a^2} \approx \sqrt{1 - \frac{1}{2}k_0^2 a^2} + \frac{mA}{\hbar k a} \cdot \frac{k_0 a}{k}$$

$$\frac{2mE}{\hbar^2} a^2 \approx 2 \frac{mA}{\hbar} + k^2 a^2$$

$$E = \frac{A\hbar}{a^2} + k^2 \frac{\hbar^2}{2m} = \frac{\hbar^2 k^2}{2m} + \text{konst.} \Rightarrow m = m^*$$

Približek prostih elektronov

e^- ravni val: $\psi = A e^{i(kx - \omega t)}$

odbiti valovi: $\psi = A' e^{i(-kx - \omega t)}$ destruktivna interferenca, če $ka \neq n\pi$

če $ka = n\pi$:

$$\left. \begin{array}{l} \psi_+ = A e^{i(kx - \omega t)} \\ \psi_- = A e^{i(-kx - \omega t)} \end{array} \right\} \psi = \psi_+ \pm \psi_- = A(e^{i(kx - \omega t)} \pm e^{i(-kx - \omega t)})$$

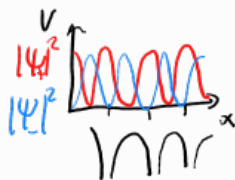
$$\rightarrow \psi_+ = 2A e^{-i\omega t} \cos kx$$

$$\psi_- = 2iA e^{-i\omega t} \sin kx$$

$$1 = \int_0^a \psi_+^* \psi_+ dx = 4A^2 \int_0^a \cos^2\left(\frac{n\pi}{a}x\right) dx = 4A^2 \cdot \frac{a}{2} = 2A^2 a \Rightarrow A^2 = \frac{1}{2a}, A = \frac{1}{\sqrt{2a}}$$

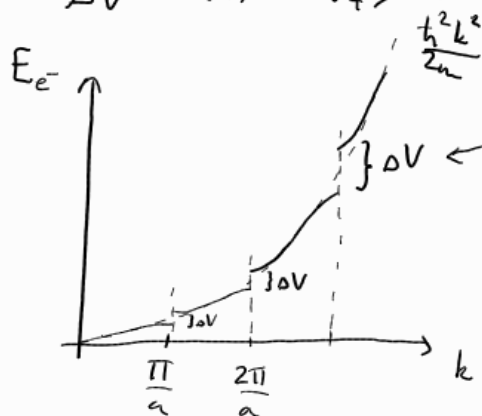
$$W_p = \langle V_+ \rangle = \int_0^a \psi_+^* V(x) \psi_+ dx < 0$$

$$\langle V_- \rangle = \dots < 0$$



Iz slike očito! $\langle V_+ \rangle < \langle V_- \rangle$

$$\Delta V = \langle V_- \rangle - \langle V_+ \rangle > 0$$



ni možno imet e^- s take E
(prevedeni pas)

Kristali v zun. el. polju

prost e^- : $v = \frac{p}{m} = \frac{\hbar k}{m}$, $W_k = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}$, $\frac{dW_k}{dk} = \frac{\hbar^2 k}{m} = \hbar v$

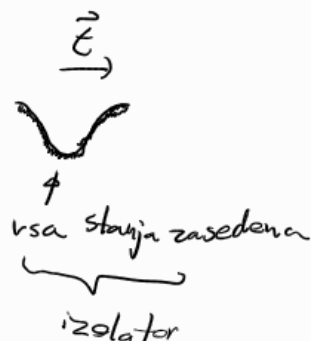
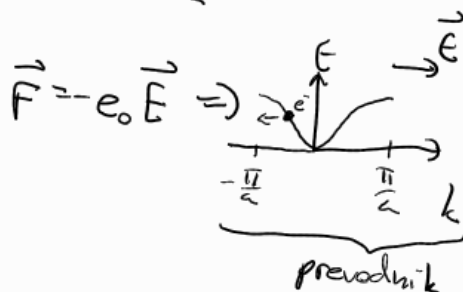
$$\rightarrow v = \frac{1}{\hbar} \frac{dW_k}{dk}$$

grupna hitrost: $\frac{dW}{dk}$

fazna hitrost: $\frac{W}{k}$

El. polje: $dW_k = F dx = F v dt = \frac{F}{\hbar} \frac{dW_k}{dk} dt$

$$\rightarrow \boxed{\frac{dk}{dt} = \frac{F}{\hbar}}$$



El. tok v kovinah

$$\frac{d\langle p \rangle}{dt} = -e_0 E - \frac{\langle p \rangle}{\tau} \rightarrow \text{STAK. STANJE: } \frac{\langle p \rangle}{\tau} = -e_0 E \xrightarrow{\langle p \rangle = m \langle v \rangle} \langle v \rangle = - \underbrace{\frac{e_0 \tau}{m}}_{\text{gibljivost } -\beta_e} E$$

$$\langle v \rangle \ll v_F$$

$$j = -e_0 n_e \langle v \rangle = \underbrace{e_0^2 n_e \tau}_{\text{prevodnost}} \frac{E}{m} = \sigma E$$

$$\bar{l} = \tau v_F \quad \text{tipična hitrost (Fermijeva)}$$

$$\bar{l} = \frac{1}{n_e \pi R^2}$$

$$\bar{l} = \frac{M \omega_D^2}{n_e \pi k_B T}$$

$$\rightarrow \tau = \frac{M \omega_D^2}{n_e \pi k_B T v_F}$$

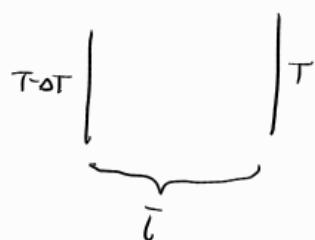
spec. upornost

$$\rho = \frac{1}{\sigma} = \frac{m n_e \pi k_B T v_F}{n_e^2 e_0^2 M \omega_D^2} = \frac{\pi m k_B T v_F}{e_0^2 M \omega_D^2} \Rightarrow \boxed{\rho \propto T}$$

SPLOŠNO (vsak atom prispeva nekaj val. e^-)

$$\boxed{\rho = \frac{\pi m n_a k_B T v_F}{e_0^2 M \omega_D^2 n_e}}$$

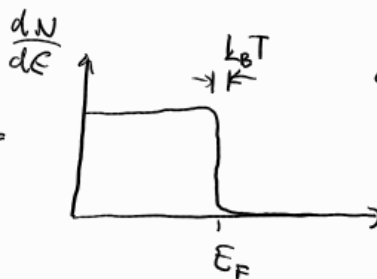
Toplotna prevodnost



$$\Delta T = \frac{dT}{dx} \bar{l} = \frac{dT}{dx} \tau v_F$$

$$j_Q = \sigma \frac{3}{2} k_B T v_F$$

gostota e^-



$$\sigma_{ne} = n_e \cdot \frac{k_B T}{E_F}$$

$$j_Q = -n_e \frac{k_B T}{E_F} \frac{3}{2} k_B \Delta T v_F = -\frac{3}{2} \frac{n_e k_B^2 T \Delta T v_F}{\frac{1}{2} m v_F^2} = -3 \frac{n_e k_B^2 T \tau}{m} \frac{dT}{dx} = -\lambda \frac{dT}{dx}$$

$$\Rightarrow \lambda = \frac{3 n_e k_B^2 T \tau}{m}$$

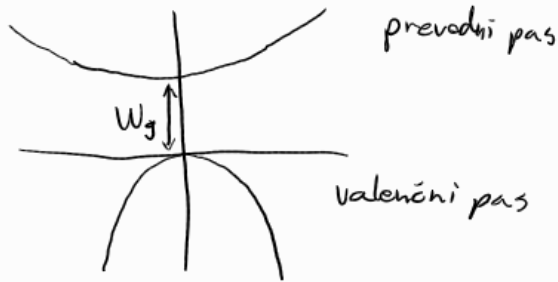
$$\frac{\lambda}{\sigma} = \frac{3 n_e k_B^2 T \tau / m}{n_e^2 e_0^2 \tau} = \frac{3 k_B^2 T}{e_0^2}$$

realen faktor $\frac{\pi^2}{3}$

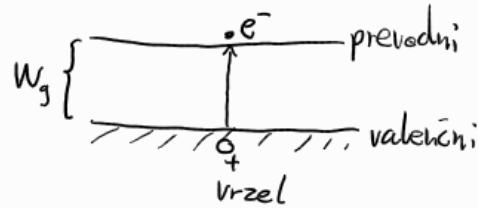
NEODV. od snovi

Polprevodnik

- izolator, ki ima špranjo (sl) velikosti $\sim 1\text{eV}$
- pri višjih T lahko e^- preskoči iz valenčnega v prevodni pas



POENOSTAVITVE:



$$W_e = W_g + \frac{1}{2} \frac{\hbar^2 k^2}{m_e^*} ; \quad W_v = -\frac{1}{2} \frac{\hbar^2 k^2}{m_v^*}$$

|
efektivna masa

• v splošnem $m_e^* \neq m_v^* \Rightarrow$ paraboli sta različno ukrivljeni

$$\rightarrow \text{el. tok } j = -e_0 n_e \langle v_e \rangle + e_0 n_v \langle v_v \rangle = e_0 n_e \beta_e E + e_0 n_v \beta_v E$$

Fermijev nivo: $W_F = \frac{1}{2} W_g$ če $m_e^* = m_v^*$ ali $T=0$
(kemijski potencial)

• dokaz/izpeljava vaje